MARKING SCHEME FOR BIOLOGY 1 - NORTHWESERN ZONE SEPT, 2024

1. 1 MARK @= 10MARKS

| i | ii | iii | iv | v | vi | vii | viii | ix | x |
|---|---|---|---|---|---|---|---|---|---|
| B | B | C | B | A | B | A | C | B | A |

2. 1 MARK @= 6MARKS

| i | ii | iii | iv | v | vi |
|---|---|---|---|---|---|
| B | D | F | K | J | H |

3. **Importance of studying biology**

- Helps to understand ourselves better
- Helps to appreciate nature
- Helps to answer fundamental questions like where life originated? Is life possible without green plants? Why black parents bear albino children? etc
- Helps to enter into different careers like teaching, medicine, etc
- Helps to understand the relationships among organisms and how to live mutually and safety
- Helps to conserve our environment for sustainable development
- Helps to understand about different diseases on their causes, symptoms, transmission, effects and prevention
- Helps to acquire different biological skills necessary for our daily life.

**(Any 6 points 1.5mark@= 9marks)**

4. a) How our bodies defend themselves against pathogens

- Skin - act as a physical barrier which protects the body against pathogens from entering the body cells and tissues
- Mucus membrane in respiratory, digestive, and urinogenital tracks helps to release mucus with antimicrobial substances that trap and deal with pathogens trying to enter the body through those areas.
- Hairs in the tubes of ears and nose help to trap dust, pathogens and any other foreign materials from entering the body via those openings. And as these foreign material hanged around, the body find a mechanism to take them out. For example through sneezing which empty the throat and the nose.
- Stomach acid (hydrochloric acid) in gastric juice released from walls of the stomach helps to kill pathogens ingested with food.

- Lysozymes which is an enzyme found in saliva, tears and sweat helps to break down bacterial cell walls to make them exposed for the other defense mechanisms to act upon.

- Phagocytes in white blood cells like neutrophils and macrophages helps to engulf and destroy pathogens.

- Inflammation is another body mechanism to defend itself against pathogens. It is a response to infection or injury that includes redness, heat, swelling and pain on the attacked area to isolate and eliminate pathogens

- Fever is also another body mechanism to defend itself against pathogens. Its response is to elevate the body temperature that helps to inhibit pathogen's growth and enhances immune function.

**(Any 5points @1 mark = 5marks)**

**b) Factors affecting body immunity**

- Lack balanced diet/ poor nutrition
- Attack by pathogens
- Incomplete treatment
- Lack of vaccination
- Extreme stress/ emotional disturbance
- Genetic disorders
- Skin damage
- Age, when young the body immunity is immature and when too old the body immunity tends to declines

**(Any 4points @1 mark = 4marks)**

5. a) The effect in recipient's blood when he/she transfused an incompatible blood from the donor is **AGGLUTINATION** **(2 mark)**

b) i). Checking blood group of the donor and the recipient
ii).Checking rhesus factor for both donor and recipient
iii).Checking blood infections of donor
**(Any 3points @1 mark = 3marks)**

c) **Components of blood and their functions**

| BLOOD COMPONENT | FUNCTION |
|---|---|
| i. Red blood cells | Transport carbon dioxide and oxygen |
| ii. White blood cells | To defend the body/Fight against diseases and infections. |
| iii. Platelets | For blood clotting |
| iv. Blood plasma | Transport Nutrients/Hormones/Waste |

**(4 components @0.5 mark= 2marks; 4functions respectively@ 0.5mark= 2marks; total = 4marks)**

6. a) **Biological and physical changes of a girl at puberty**

(i) Ovulation starts

(ii) Menstruation starts

(iii) Breasts enlarge

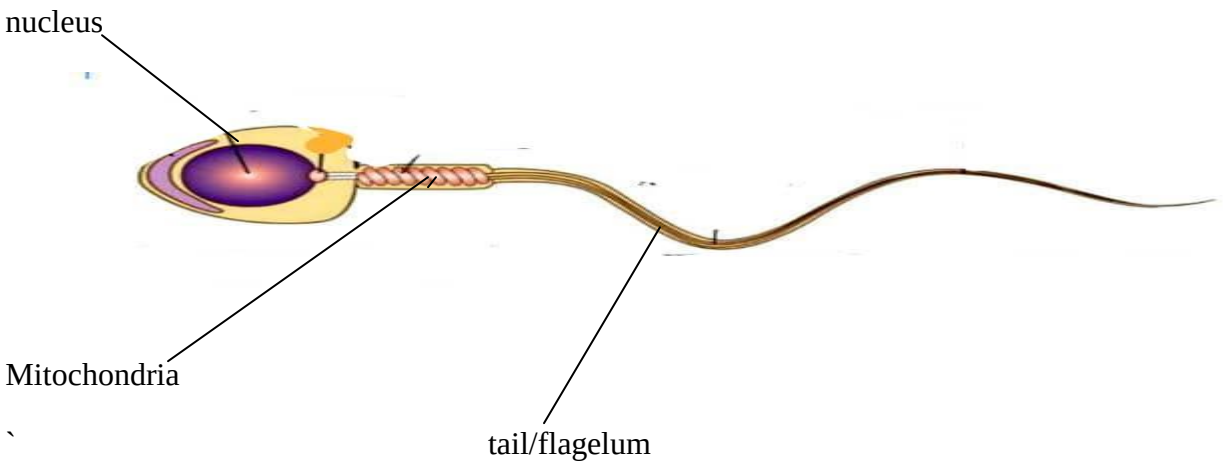
(iv) Development of soft and high pitched voice

(v) Broadening of hips

vi) Enlargement of vagina and vulva.

**(Any 4points @1 mark = 4marks)**

b) A Diagram of Male sex cell (sperm)




**A Diagram of Male sex cell (sperm)**

**Diagram 1mark**
**Labeling 3parts @1mark =3marks**
**Caption 1mark**
} **total 5marks**

7. (a) Multiple pregnancies. **(01 mark)**

(b) **Causes of multiple pregnancy**.

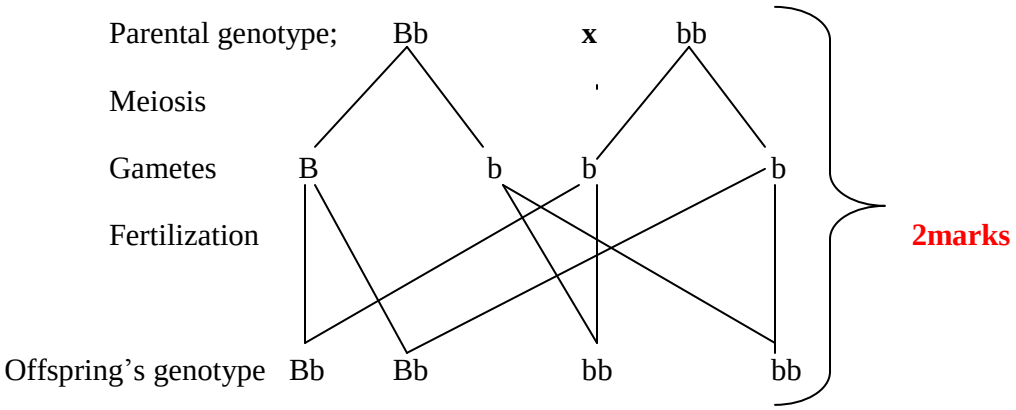
- **Genetic factors**- A family with history of twins have a higher chance of having twins.
- **Use of fertility drugs**. These cause multiple ovulation thus increasing the chances of multiple pregnancy.
- **In Vitro fertilization**: These are assisted reproductive technologies that help woman to conceive. It involves introducing zygote formed outside the body to the uterus of a female.

**(Any2 points @1mark =2marks)**

(c) **Differences between identical and fraternal twins**. **(3points @2marks= 6marks)**

| **Identical twins** | **Fraternal twins** |
|---|---|
| (i) Result form the same ovum (egg) fertilized by the same sperm. | Result from different ovum (eggs) fertilized by different sperms. |
| (ii) They look alike in almost all aspects and usually of the same sex. | They don't necessarily look alike and can have different sex |
| (iii) Can share the same placenta and amniotic sac. | Do not share the same placenta and amniotic sac. |

8. i)Let B = black gene colour
b = white gene colour } **0.5 mark**

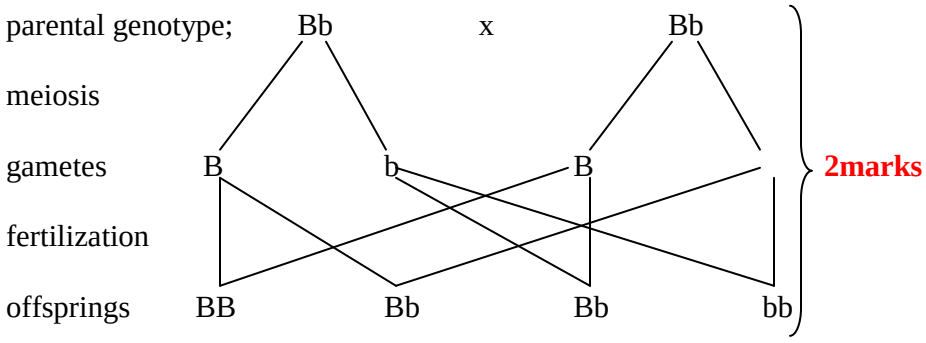



The phenotypes of the offsprings= 2 black
2 white } **0.5mark**

Hence, 2/4X100 = 50% Black
2/4X100 = 50% White } **0.5 mark**

Therefore, the parents should be heterozygous black and homozygous white to produce 50%black and 50%white. **(1mark)**

ii). Let parental phenotype be black x black **(0.5mark)**




The phenotypes of the offsprings= 3 black
1 white **0.5mark**

Hence, 3/4X100 = 75% Black
2/4X100 = 25% White **0.5 mark**

**Therefore, the parents should both be heterozygous black to produce 75% black and 25 white (1mark)**

**SECTION C; ESSAY TYPE QUESTIONS**

9. Introduction –meaning of breast feeding **(2 marks)**
Main body –Importance of breastfeeding
   - Colostrums provide antibodies for baby's immunity
   - It has right temperature
   - It is a balanced diet (possesses all nutrients required by the baby)
   - It is safe and free from contamination
   - Needs no preparations, always there when required
   - Breast feeding builds emotional bond between mother and baby
   - Breast feeding reduces the ability of the mother to conceive early

   **(any 6 points 2@= 12marks)**

**Any related conclusion (1 mark)**

10. **Introduction -** Meaning of movement **(02marka)**

**Main body; Adaptation of muscles to movement**

- Muscles have many mitochondria for energy production in form of ATP
- Muscles are multinucleated for better coordination to its contractile activities
- Muscles are elastic to allow contraction and relaxation
- Muscles are highly branched to provide large surface area for contraction and relaxation
- Muscles are highly supplied with blood capillaries to supply nutrients and oxygen
- Cardiac muscles are myogenic that contract and relax for long time without becoming fatigue.

  **(Any three points @02 = 6marks)**

**Significances of Tropism**

- Tropism help plant leaves/shoots to move towards light for photosynthesis to take place (phototropism)
- Tropism help plant roots to move towards water and mineral salts for photosynthesis (hydrotropism)
- Tropism help plant stem and root to move toward gravity and provide upright (geotropism)
- Tropism help plant to move towards or away from chemicals(chemotropism)

**(Any three points @02 = 6marks)**

**Any related conclusion (1 mark)**

11. **Introduction -** Meaning of Communicable diseases **(02marks)**

Main body; **ways through which communicable diseases are transmitted**;

- Through air. Eg flue, turberculosis, $COVID$ 19
- Through sexual intercourse. Eg HIV/AIDS, gonorrhea, syphilis, candidiasis etc
- Through contaminated water and food. Eg cholera and typhoid
- Through blood transfusion. Eg HIV/AIDS, malaria, hepatitis b etc
- Through vector. Eg malaria, sleeping sickness, plague etc
- Through contact. Eg $COVID$ 19, scabies, ringworms etc

**(Naming a way 1mark, explanation 1mark, 2 examples 0.5 @=1mark =3marks x 4points =12marks)**

**Any related conclusion (1 mark)**